

Chapter 1

Subject Matter Overview

A **control system** is an interconnection of devices that alter the natural behavior of a **plant**. A plant is itself a device we aim to manipulate. Some basic examples should help to clarify the definitions

1. Rocket During Launch

Plant: Rocket; which consists of mechanical, electrical, electronic, and chemical systems

Control system: sensors that measure the rocket's position and attitude; effectors that manipulate the rocket engine nozzle

- (a) Natural behavior (without control): the forces of thrust and gravity to align driving the rocket into the ground
- (b) Controlled behavior: align the thrust of the rocket in a direction to propel the rocket into space

2. Home Oven

Plant: Oven, which consists of electronic and chemical (the fuel) systems

Control system: thermostat, flow control effector on the fuel line

- (a) Natural behavior: heat the volume to maximum temperate possible
- (b) Controlled behavior: heat to a set temperature and maintain that temperature for cooking

3. Self-Balancing Bicycle

Plant: Bicycle; which consists of mechanical systems

Control system: sensors that monitor the tip angle of the of plant; effectors that control the steering angle of the front tire

- (a) Natural behavior: the bicycle falls over
- (b) Controlled behavior: balance the vehicle as it is propelled forward

For the three examples above, and for most systems, accurate mathematical models that represent the physical characteristics of the system and the controls devices are necessary. Note that for the rocket and the bicycle without control the natural response will lead to *detrimental results*; without proper control, the outcome results in damage to the plant and likely the control system. These systems are considered **unstable** because the uncontrolled result damages the plant and produces an unbounded result (mathematical definition) – much more on this later.

The study of control systems may be separated into three areas: **model development**, **analysis**, and **controller design**. In model development, we find new ways to mathematically model systems we want to understand and manipulate. In analysis, we investigate a system with proven techniques. Finally, for controller design we design and implement a controller to manipulate a system to achieve our controls objectives. The time to go from “idea” to final design may be weeks or, for complex problems, years. This content touches on all three aspects, but focuses on analysis and controller design.

Generally, differential equations are transformed into Laplace domain equivalent expressions where they are manipulated into standard forms such that the dynamic response can be determined without actually needing to perform time domain simulation. In ECE 4375 the models are linear time-invariant which allows for highly accurate model development. For more challenging systems, such as nonlinear and time-varying devices, advanced techniques are employed — they’re just beyond the scope of a senior level course. Modern approaches use software simulation tools, such as MATLAB, to directly observe the time domain behavior of the system.

Conversion from the time domain differential equation, such as

$$\ddot{y}(t) + a_1\dot{y}(t) + a_0y(t) - \dot{u}(t) - b_0u(t) = 0, \quad (1.1)$$

to the Laplace domain equivalent equation is typically performed with a lookup table.

With the case of (1.1), the Laplace equivalent equation is

$$s^2Y(s) - sy(0) - \dot{y}(0) + a_1sY(s) - a_1y(0) + a_0Y(s) - sU(s) + u(0) - b_0U(s) = 0 \quad (1.2)$$

where the $y(0)$ and $u(0)$ terms are the initial conditions. For many studies – including this course – the initial conditions are assumed zero. Therefore, (1.2) simplifies to

$$s^2Y(s) + a_1sY(s) + a_0Y(s) - sU(s) - b_0U(s) = 0 \quad (1.3)$$

Now, in the Laplace form, the system is manipulated into a transfer function¹. A transfer function is an equation that is the ratio of output to input. In this class, the system output will always be $y(t)$ in the time domain and $Y(s)$ in the Laplace domain. The input will be either $u(t)$, $U(s)$ or $r(t)$, $R(s)$. In the case of (1.3), the input is $U(s)$, the output is $Y(s)$. To re-write the equation into a transfer function, first move the input expressions to the right

$$s^2Y(s) + a_1sY(s) + a_0Y(s) = sU(s) + b_0U(s) \quad (1.4)$$

factor the input and output from the right and left of the equation

$$(s^2 + a_1s + a_0)Y(s) = (s + b_0)U(s)$$

¹When the end goal is a transfer function all initial conditions are zero, $y(0) = u(0) = 0$.

Next, divide both sides of the equation by the left-hand expression

$$Y(s) = \frac{(s + b_0)U(s)}{(s^2 + a_1s + a_0)}$$

Finally, divide both sides by the input signal to create the transfer function, which is a output over input ratio

$$\frac{Y(s)}{U(s)} = \frac{s + b_0}{s^2 + a_1s + a_0} \quad (1.5)$$

Now, (1.5) is the Laplace Domain transfer function of (1.1). They are equivalent equations of the same mathematical system. We may transform from the time domain to the Laplace domain and back to the time domain without loss of information, knowledge, or generality. This statement is true for all time domain, Laplace domain representations.

In industry, many devices have a datasheet with a transfer function. When this is the case, the mathematical models may be cascaded together to form an overall model of the system, which then may be analyzed to understand the performance of the system before ever physically constructing the plant. For example, a complex robot consists of electronic devices (e.g. amplifiers and sensors), mechanical members, and actuators (e.g. DC-servo motors). The systems are arranged such that the robotic device performs as designed.

In general, most systems, regardless of complexity, may be mathematically modeled. Of course, more complex the system (e.g. nonlinear and/or time-varying) the more difficult it is to establish a mathematical model.

In this course, the focus is on linear time-invariant systems with little or no noise which allows for a near perfect mathematical model. There are some linear devices that are common and reoccur in most non-trivial systems.

1.1 Basic Models

Let's take a look at a couple of math models:

1. Amplifier

Input, U: low-voltage, low-power

Output, Y: higher-voltage, higher-power

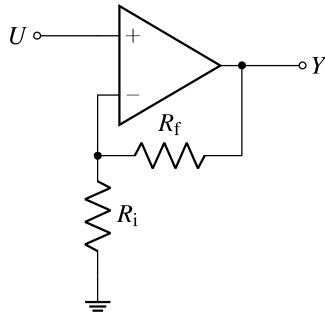
Device: operational amplifier (op amp)

The transfer function

$$\frac{Y(s)}{U(s)} = K_{\text{oa}} \quad (1.6)$$

where $-\infty < K_{\text{oa}} < \infty$.

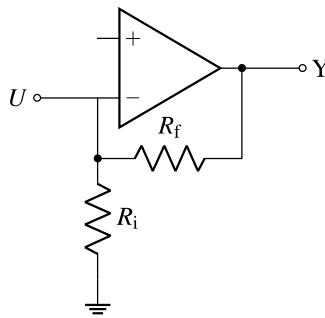
For example, if we use an op amp circuit



then

$$K_{\text{oa}} = 1 + \frac{R_f}{R_i} = \frac{R_i + R_f}{R_i}$$

If we apply the input to the inverting terminal, such as



then

$$K_{\text{oa}} = -\frac{R_f}{R_i}$$

2. Summation Junction

Input, U_i : Multiple signal sources

Output, Y: summation of signal sources

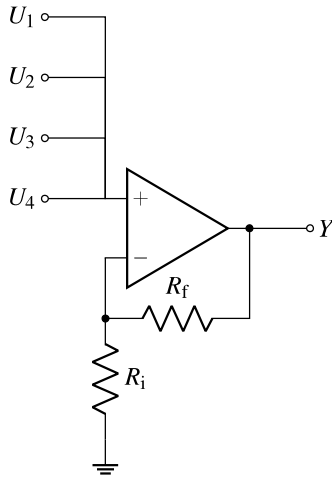
Device: operational amplifier (op amp)

Mathematical representation:

$$Y(s) = \sum_{i=1}^n U_i(s) \quad (1.7)$$

Note: the summation junction is not a transfer function

The summation junction circuit with four inputs



here

$$\begin{aligned}
 Y &= \left(1 + \frac{R_f}{R_i}\right)(U_1 + U_2 + U_3 + U_4) \\
 &= \left(\frac{R_i + R_f}{R_i}\right)(U_1 + U_2 + U_3 + U_4)
 \end{aligned}$$

1.2 Time Domain to Laplace Domain Transformations

Transfer functions are the primary modeling tool we use for this course – it's also a incredibly common in industry. When possible, we chose to work with transfer functions.

Transfer functions require a Laplace domain representation. When given a time domain equation, we use a Look Up Table, LUT, to transform between the time and Laplace domains. If the provided table is insufficient to complete a transformation, find another table. There is no expectation to use the mathematical equations to complete a transformation. However, no set of notes is complete without the equations

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

and

$$f(t) = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds$$

for $t > 0$ and where c is larger than the real parts of all singular points of $F(s)$.

1.3 Selected Functions

Some functions are more common in the field than others. Below are the most common input signals and the exponential function.

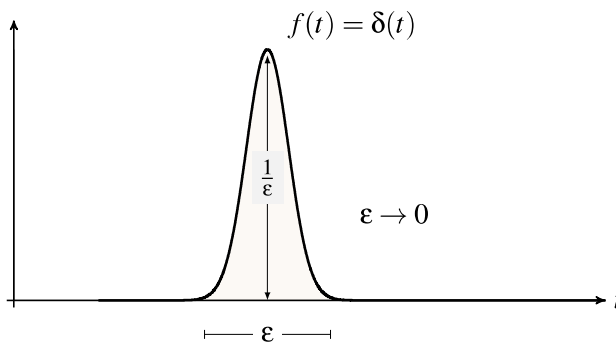
	$f(t)$	$F(s)$	Notes
1	$k \cdot f(t)$	$kF(s)$	k is a constant
2	$f_1(t) + f_2(t)$	$F_1(s) + F_2(s)$	
3	$\delta(t)$	1	impulse
4	$1(t)$	$\frac{1}{s}$	step
5	t	$\frac{1}{s^2}$	
6	$\frac{t^2}{2}$	$\frac{1}{s^3}$	
7	t^n	$\frac{n!}{s^{n+1}}$	$n = 1, 2, 3, \dots$
8	$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$	$n = 1, 2, 3, \dots$
9	e^{-at}	$\frac{1}{s+a}$	
10	$\sin(bt)$	$\frac{b}{s^2+b^2}$	
11	$\cos(bt)$	$\frac{s}{s^2+b^2}$	
12	$e^{-at} \sin(bt)$	$\frac{b}{(s+a)^2+b^2}$	
13	$e^{-at} \cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$	
14	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$	$n = 1, 2, 3, \dots$
15	$\dot{f}(t)$	$sF(s) - f(0)$	$f(0)$ is initial condition
16	$\frac{d^n f(t)}{dt^n}$	$s^n F(s) + \text{initial conditions}$	initial conditions are zero in ECE 4375
17	$\int f(t) dt$	$\frac{F(s)}{s}$	
18	$\int \dots \int f(t) dt$	$\frac{F(s)}{s^n}$	n integrations
19	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$	convolution
20	$f_1(t) \cdot f_2(t)$	$F_1(s) * F_2(s)$	convolution
21	$\lim_{t \rightarrow 0} f(t)$	$\lim_{s \rightarrow \infty} sF(s)$	IVT
22	$\lim_{t \rightarrow \infty} f(t)$	$\lim_{s \rightarrow 0} sF(s)$	FVT

Impulse Function

While we call the impulse a function, it's technically not a function because of the infinite height and zero width. It is however a useful tool in software simulation and mathematical analysis.

$$f(t) = \delta(t)$$

which produces the plot



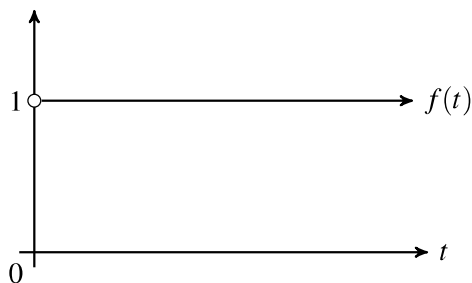
Where the area under the curve is one.

Step Function

The step function serves as a setpoint input, which unlike the impulse, is physically realizable. It is a bounded, static input

$$f(t) = \begin{cases} 1, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

which produces the plot



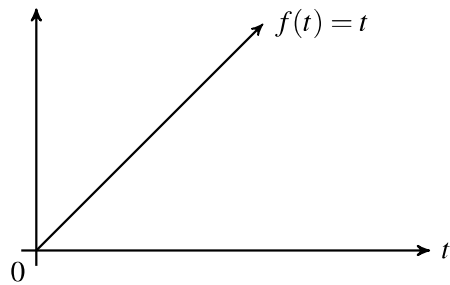
Think of the temperature setting for a home; we adjust the temperature setting on the thermostat which functions as the setpoint input into the temperature control system.

Ramp Function

The ramp function is an unbounded input

$$f(t) = t$$

used to determine the capacity of a system to follow an input with a constant velocity.

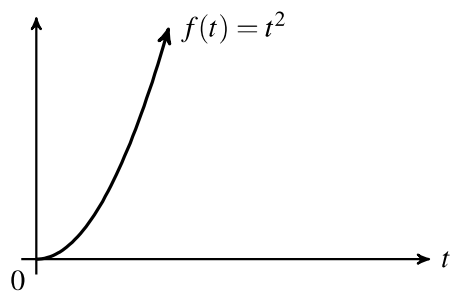


Parabola Function

The parabola function is an unbounded input

$$f(t) = t^2$$

used to determine the capacity of a system to follow an input with a constant acceleration.

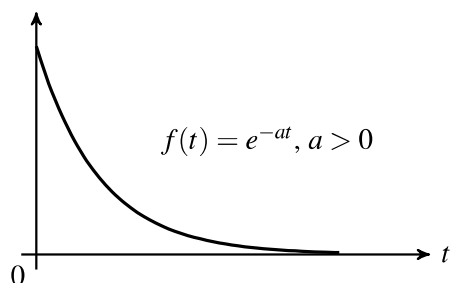


Exponential Function

We see the exponential function not as a signal input, but as an output signal of a transfer function.

$$f(t) = e^{-at}$$

The stability of exponential function is determined by the constant **a**. When **a** is positive, the system is stable and the time-domain response is



When **a** is negative, the system is unstable and the time-domain response is

